Τραπεζικό μάρκετινγκ (bank.csv) — αναφορά εξόρυξης δεδομένων

Weka Web · Bank Marketing WEKA Walkthrough Πηγή: bank.csv 4521 εγγραφές · 16 χαρακτηριστικά

1. Σύνοψη & συμπεράσματα

Η αναφορά ακολουθεί το «Bank Marketing (bank.csv) WEKA Walkthrough». Τα δεδομένα έχουν 4.521 πελάτες και 17 χαρακτηριστικά· μόνο 521 πελάτες (11,5%) έκαναν την προθεσμιακή κατάθεση. Ένα μοντέλο που απαντά πάντα «όχι» έχει ήδη ακρίβεια 88,5% χωρίς να βρίσκει κανέναν συνδρομητή, οπότε τα μοντέλα κρίνονται με precision και recall για το «yes», την περιοχή ROC και το επιχειρηματικό κόστος των λαθών (€50 για κάθε χαμένο συνδρομητή, €2 για κάθε άσκοπη κλήση).

Διαρροή δεδομένων. Το InfoGain κατατάσσει πρώτο το «duration» (0,108 — τριπλάσιο από το επόμενο, το poutcome με 0,038) και τα μοντέλα της Έκδοσης A διασπούν αμέσως με αυτό (βλ. κανόνες JRip και δέντρο J48 στο παράρτημα). Η διάρκεια της κλήσης είναι γνωστή μόνο μετά την κλήση, άρα η Έκδοση A (ακρίβεια 86,9–90,2%) είναι μέτρο σύγκρισης και όχι μοντέλο που μπορεί να χρησιμοποιήσει η τράπεζα.

Η Έκδοση B (χωρίς duration) είναι το ρεαλιστικό μοντέλο. Η ακρίβεια μένει στο 83,0–89,2%, αλλά το recall για το «yes» πέφτει στο 0,14–0,36 και η περιοχή ROC στο 0,58–0,72. Τα πιο χρήσιμα σήματα που απομένουν είναι το αποτέλεσμα της προηγούμενης καμπάνιας (το 64% όσων έχουν poutcome = success κάνει κατάθεση), οι ημέρες από την τελευταία επαφή και ο μήνας. Το J48 γίνεται δέντρο τεσσάρων φύλλων πάνω στο poutcome. Το νέο χαρακτηριστικό contact_ratio = campaign / (previous + 1) έχει InfoGain 0,017, περισσότερο από τα campaign (0,003) ή previous (0,016) μόνα τους.

Επιχειρηματική απόφαση. Με αυτά τα κόστη, το NaiveBayes έχει το χαμηλότερο αναμενόμενο κόστος (€17.520: 333 χαμένοι συνδρομητές και 435 άσκοπες κλήσεις), παρότι έχει τη χαμηλότερη ακρίβεια (83,0%). Τα J48, Logistic και JRip χάνουν 429–448 συνδρομητές (€21.570–€22.498), ενώ το «πάντα όχι» θα κόστιζε €26.050. Πρόταση: χρήση του NaiveBayes (ή του Logistic με χαμηλότερο κατώφλι πιθανότητας) για τη λίστα κλήσεων, με επανεξέταση του κατωφλίου όταν αλλάζουν τα κόστη.

Τμήματα πελατών (k-means, k = 4, μόνο χαρακτηριστικά προφίλ — το y και τα στοιχεία καμπάνιας αγνοήθηκαν). Το τμήμα 0 (319 πελάτες· περίπου 50 ετών, ελεύθεροι επαγγελματίες, τριτοβάθμια εκπαίδευση, χωρίς στεγαστικό) έχει μετατροπή 19,4% και το τμήμα 2 (1.382 πελάτες· περίπου 42, στελέχη, έγγαμοι, τριτοβάθμια, χωρίς στεγαστικό) 13,5%, πάνω από τον μέσο όρο 11,5%. Το τμήμα 3 (1.589 πελάτες· περίπου 36, εργάτες, έγγαμοι, με στεγαστικό) μόλις 8,5%. Το EM, που επιλέγει τον αριθμό ομάδων με cross-validation, προτείνει 8 μικρότερα τμήματα. Πρόταση: προτεραιότητα σε μεγαλύτερους και πτυχιούχους πελάτες χωρίς στεγαστικό δάνειο και χαμηλότερη προτεραιότητα στους νέους εργάτες με στεγαστικό το επόμενο τρίμηνο.

2. Σύνολο δεδομένων

Το σύνολο δεδομένων όπως είναι μετά τα βήματα προεπεξεργασίας παρακάτω.

ΣΧΕΣΗ	ΕΓΓΡΑΦΕΣ	ΧΑΡΑΚΤΗΡΙΣΤΙΚΑ	ΕΛΛΙΠΕΙΣ ΤΙΜΕΣ	ΚΛΑΣΗ
bank	**4.521**	**16**	**0%**	**y**

Κατανομή της κλάσης «y»



#	Χαρακτηριστικό	Τύπος	Ελλιπείς	Διακριτές	Στατιστικά / συχνότερες τιμές
1	**age**	numeric	0	67	min 19 · max 87 · μέσος 41.17 · sd 10.58
2	**job**	nominal	0	12	management (969), blue-collar (946), technician (768), admin. (478) …
3	**marital**	nominal	0	3	married (2797), single (1196), divorced (528)
4	**education**	nominal	0	4	secondary (2306), tertiary (1350), primary (678), unknown (187)
5	**default**	nominal	0	2	no (4445), yes (76)
6	**balance**	numeric	0	2353	min -3313 · max 71188 · μέσος 1422.66 · sd 3009.64
7	**housing**	nominal	0	2	yes (2559), no (1962)
8	**loan**	nominal	0	2	no (3830), yes (691)
9	**contact**	nominal	0	3	cellular (2896), unknown (1324), telephone (301)
10	**day**	numeric	0	31	min 1 · max 31 · μέσος 15.92 · sd 8.25
11	**month**	nominal	0	12	may (1398), jul (706), aug (633), jun (531) …
12	**campaign**	numeric	0	32	min 1 · max 50 · μέσος 2.79 · sd 3.11
13	**pdays**	numeric	0	292	min -1 · max 871 · μέσος 39.77 · sd 100.12
14	**previous**	numeric	0	24	min 0 · max 25 · μέσος 0.54 · sd 1.69
15	**poutcome**	nominal	0	4	unknown (3705), failure (490), other (197), success (129)
16	**y** κλάση	nominal	0	2	no (4000), yes (521)

3. Βήματα προεπεξεργασίας

1. Φόρτωση bank.csv 01:56:53 π.μ.
2. Αφαίρεση χαρακτηριστικών: duration 01:56:56 π.μ.

4. Ταξινόμηση

Σύγκριση μοντέλων

Θετική κλάση «yes». Μοντέλο κόστους: ένα χαμένο «yes» (ψευδώς αρνητικό) κοστίζει €50, ένας ψευδής συναγερμός (ψευδώς θετικό) €2.

#	Μοντέλο	Δεδομένα	Ακρίβεια	Kappa	Precision (yes)	Recall (yes)	F (yes)	ROC AUC	FN	FP	Κόστος (€)
1	**J48** 10-fold CV	bank · 17 χαρ.	88.98%	0.368	0.533	0.355	0.426	0.762	336	162	€17.124
2	**NaiveBayes** 10-fold CV	bank · 17 χαρ.	86.88%	0.398	0.44	**0.509**	**0.472**	0.845	256	337	**€13.474**
3	**Logistic** 10-fold CV	bank · 17 χαρ.	**90.18%**	0.386	0.646	0.326	0.434	**0.888**	351	93	€17.736
4	**JRip** 10-fold CV	bank · 17 χαρ.	89.74%	0.387	0.593	0.349	0.44	0.67	339	125	€17.200
5	**J48** 10-fold CV	bank · 16 χαρ.	89.01%	0.192	0.598	0.14	0.227	0.6	448	49	€22.498
6	**NaiveBayes** 10-fold CV	bank · 16 χαρ.	83.01%	0.232	0.302	0.361	0.329	0.715	333	435	€17.520
7	**Logistic** 10-fold CV	bank · 16 χαρ.	89.14%	0.205	0.619	0.15	0.241	0.723	443	48	€22.246
8	**JRip** 10-fold CV	bank · 16 χαρ.	89.18%	0.234	0.605	0.177	0.273	0.579	429	60	€21.570

1. J48 10-fold cross-validation

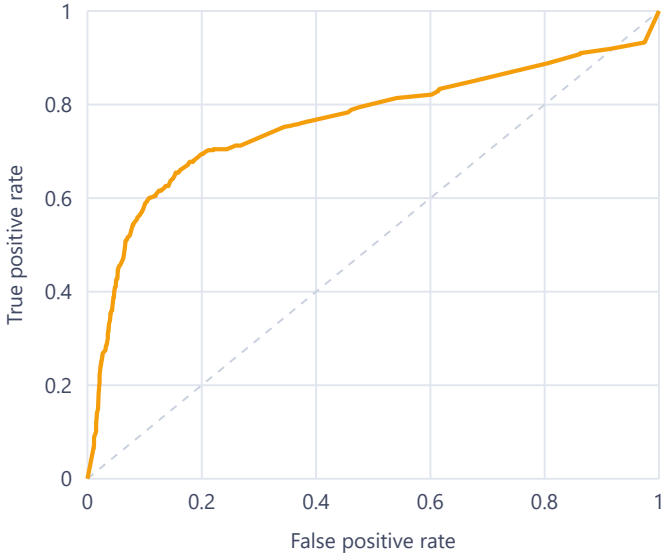
weka.classifiers.trees.J48 -C 0.25 -M 2 · bank (17 χαρακτηριστικά)

ΑΚΡΙΒΕΙΑ	ΚΑΠΠΑ	MAE	RMSE	ROC AUC (ΣΤΑΘΜ.)
88.98%	**0.368**	**0.1448**	**0.2977**	**0.762**

Πίνακας σύγχυσης (γραμμές = πραγματική)

Πραγματική ↓ \ Πρόβλεψη →	no	yes
no	3838	162
yes	336	185

Καμπύλη ROC για «yes» (AUC 0.762)



Κλάση	TP rate	FP rate	Precision	Recall	F-Measure	MCC	ROC	PRC
no	0.96	0.645	0.92	0.96	0.939	0.377	0.762	0.932
yes	0.355	0.041	0.533	0.355	0.426	0.377	0.762	0.405

Μοντέλο

```
J48 pruned tree
------------------

duration <= 211: no (2548.0/73.0)
duration > 211
|   duration <= 645
|   |   poutcome = unknown
|   |   |   age <= 60
|   |   |   |   contact = cellular
|   |   |   |   |   month = oct
|   |   |   |   |   |   balance <= 2469: yes (7.0)
|   |   |   |   |   |   balance > 2469: no (5.0/1.0)
|   |   |   |   |   month = may: no (102.0/14.0)
|   |   |   |   |   month = apr
|   |   |   |   |   |   day <= 20: no (58.0/5.0)
|   |   |   |   |   |   day > 20
|   |   |   |   |   |   |   duration <= 238: no (3.0)
|   |   |   |   |   |   |   duration > 238: yes (13.0/1.0)
|   |   |   |   |   month = jun: yes (21.0/7.0)
|   |   |   |   |   month = feb
|   |   |   |   |   |   day <= 7
|   |   |   |   |   |   |   balance <= 1: yes (3.0/1.0)
|   |   |   |   |   |   |   balance > 1: no (37.0)
|   |   |   |   |   |   day > 7: yes (9.0/3.0)
|   |   |   |   |   month = aug: no (153.0/19.0)
|   |   |   |   |   month = jan: no (32.0/1.0)
|   |   |   |   |   month = jul: no (192.0/7.0)
|   |   |   |   |   month = nov: no (86.0/9.0)
|   |   |   |   |   month = sep: no (7.0/1.0)
|   |   |   |   |   month = mar
|   |   |   |   |   |   housing = no
|   |   |   |   |   |   |   duration <= 312: no (2.0)
|   |   |   |   |   |   |   duration > 312: yes (2.0)
|   |   |   |   |   |   housing = yes: yes (4.0)
|   |   |   |   |   month = dec: no (3.0/1.0)
|   |   |   |   contact = unknown: no (464.0/16.0)
|   |   |   |   contact = telephone: no (50.0/10.0)
|   |   |   age > 60
|   |   |   |   age <= 68: yes (21.0/8.0)
|   |   |   |   age > 68: no (16.0/7.0)
|   |   poutcome = failure
|   |   |   pdays <= 373: no (163.0/26.0)
|   |   |   pdays > 373
|   |   |   |   balance <= 2581: yes (9.0)
|   |   |   |   balance > 2581: no (2.0)
… (108 ακόμη γραμμές — δες το παράρτημα)
```

2. NaiveBayes

10-fold cross-validation

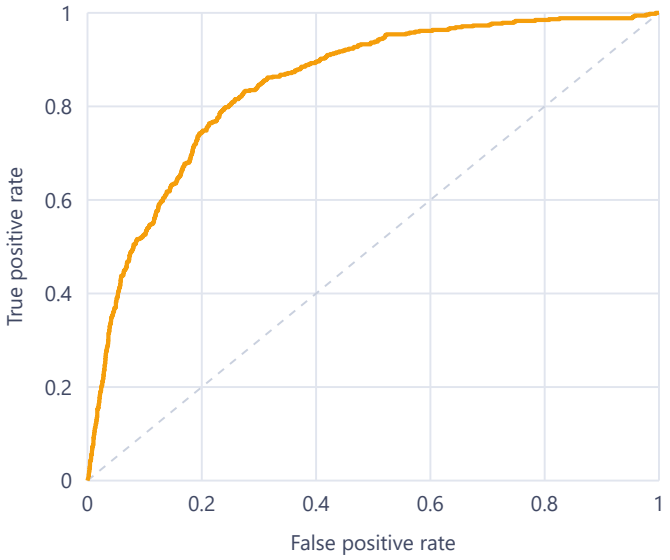
weka.classifiers.bayes.NaiveBayes · bank (17 χαρακτηριστικά)

ΑΚΡΙΒΕΙΑ	ΚΑΡΡΑ	MAE	RMSE	ROC AUC (ΣΤΑΘΜ.)
86.88%	**0.398**	**0.1625**	**0.3233**	**0.845**

Πίνακας σύγχυσης (γραμμές = πραγματική)

Πραγματική ↓ \ Πρόβλεψη →	no	yes
no	3663	337
yes	256	265

Καμπύλη ROC για «yes» (AUC 0.845)



Κλάση	TP rate	FP rate	Precision	Recall	F-Measure	MCC	ROC	PRC
no	0.916	0.491	0.935	0.916	0.925	0.399	0.845	0.972
yes	0.509	0.084	0.44	0.509	0.472	0.399	0.845	0.405

Μοντέλο

```
Naive Bayes Classifier

                      no     yes
Attribute          (0.88)  (0.12)
================================
age
  mean            41.0264 42.5015
  std. dev.       10.2053  13.145
  weight sum         4000     521
  precision        1.0303  1.0303

job
  unemployed          116      14
  services            380      39
  management          839     132
  blue-collar         878      70
  self-employed       164      21
  technician          686      84
  entrepreneur        154      16
  admin.              421      59
  student              66      20
  housemaid            99      15
  retired             177      55
  unknown              32       8
  [total]            4012     533

marital
  married            2521     278
  single             1030     168
  divorced            452      78
  [total]            4003     524

education
  primary             615      65
  secondary          2062     246
  tertiary           1158     194
  unknown             169      20
  [total]            4004     525

default
  no                 3934     513
  yes                  68      10
  [total]            4002     523

balance
… (73 ακόμη γραμμές — δες το παράρτημα)
```

3. Logistic 10-fold cross-validation

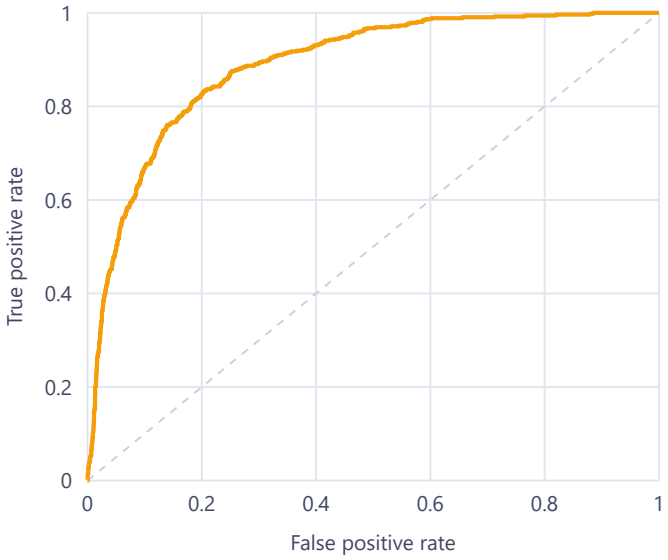
weka.classifiers.functions.Logistic -R 1e-8 -M -1 · bank (17 χαρακτηριστικά)

ΑΚΡΙΒΕΙΑ	ΚΑΡΡΑ	MAE	RMSE	ROC AUC (ΣΤΑΘΜ.)
90.18%	**0.386**	**0.1422**	**0.2723**	**0.888**

Πίνακας σύγχυσης (γραμμές = πραγματική)

Πραγματική ↓ \ Πρόβλεψη →	no	yes
no	3907	93
yes	351	170

Καμπύλη ROC για «yes» (AUC 0.888)



Κλάση	TP rate	FP rate	Precision	Recall	F-Measure	MCC	ROC	PRC
no	0.977	0.674	0.918	0.977	0.946	0.413	0.888	0.983
yes	0.326	0.023	0.646	0.326	0.434	0.413	0.888	0.506

Μοντέλο

```
Logistic Regression with ridge parameter of 1.0E-8
Coefficients...
                           Class
Variable                      no
================================
age                         0.0042
job=unemployed              0.5026
job=services                0.0088
job=management             -0.0639
job=blue-collar             0.2555
job=self-employed           0.0443
job=technician              0.0557
job=entrepreneur            0.1129
job=admin.                 -0.1369
job=student                -0.5153
job=housemaid               0.2161
job=retired                -0.7684
job=unknown                -0.6576
marital=married             0.1507
marital=single             -0.0137
marital=divorced           -0.3188
education=primary           0.1125
education=secondary         0.0324
education=tertiary         -0.2082
education=unknown           0.5336
default                    -0.5446
balance                          0
housing                       0.26
loan                        0.6296
contact=cellular           -0.5959
contact=unknown             0.8202
contact=telephone          -0.5257
day                        -0.0164
month=oct                  -1.6355
month=may                   0.2155
month=apr                  -0.2745
month=jun                  -0.8287
month=feb                  -0.4767
month=aug                   0.0337
month=jan                   0.8488
month=jul                    0.477
month=nov                   0.5685
month=sep                  -0.9317
month=mar                   -1.773
month=dec                  -0.3889
… (64 ακόμη γραμμές — δες το παράρτημα)
```

4. JRip

10-fold cross-validation

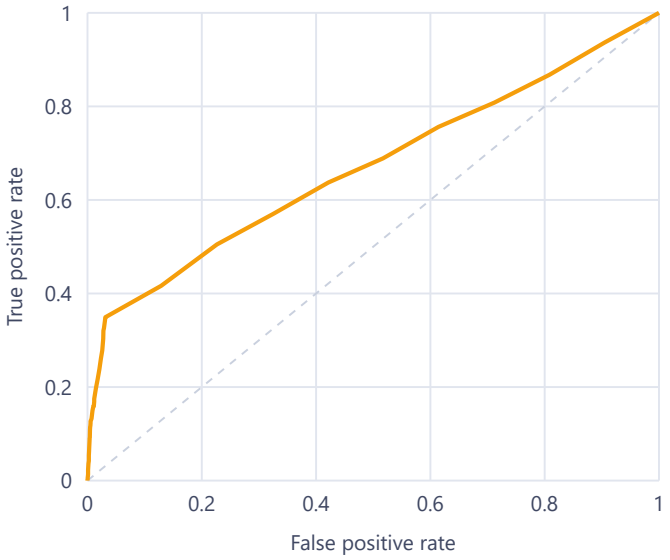
weka.classifiers.rules.JRip -F 3 -N 2 -O 2 -S 1 · bank (17 χαρακτηριστικά)

ΑΚΡΙΒΕΙΑ	ΚΑΡΡΑ	MAE	RMSE	ROC AUC (ΣΤΑΘΜ.)
89.74%	**0.387**	**0.1651**	**0.2916**	**0.67**

Πίνακας σύγχυσης (γραμμές = πραγματική)

Πραγματική ↓	Πρόβλεψη → no	yes
no	3875	125
yes	339	182

Καμπύλη ROC για «yes» (AUC 0.67)



Κλάση	TP rate	FP rate	Precision	Recall	F-Measure	MCC	ROC	PRC
no	0.969	0.651	0.92	0.969	0.944	0.404	0.67	0.925
yes	0.349	0.031	0.593	0.349	0.44	0.404	0.67	0.358

Μοντέλο

```
JRIP rules:
===========

(duration >= 651) and (contact = cellular) and (campaign >= 4) and (loan = no) => y=yes (40.0/10.0)
(duration >= 757) => y=yes (207.0/98.0)
(duration >= 212) and (pdays >= 56) and (poutcome = success) => y=yes (82.0/19.0)
(duration >= 212) and (contact = cellular) and (pdays >= 375) => y=yes (11.0/1.0)
(duration >= 342) and (housing = no) and (month = oct) => y=yes (14.0/3.0)
 => y=no (4167.0/298.0)

Number of Rules : 6
```

5. J48

10-fold cross-validation

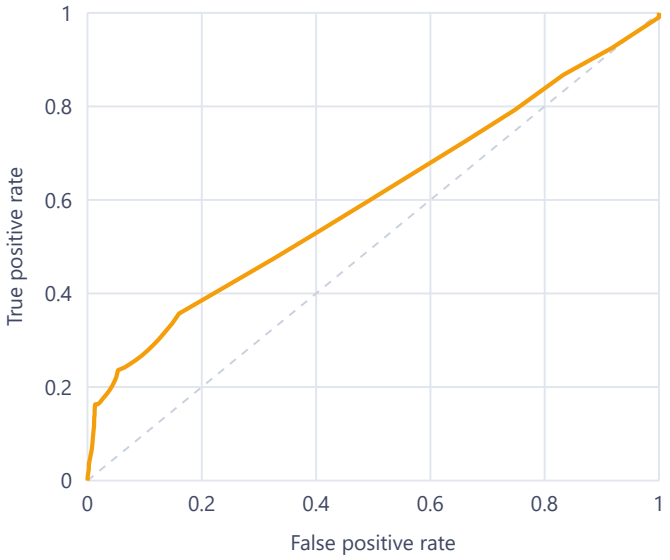
weka.classifiers.trees.J48 -C 0.25 -M 2 · bank (16 χαρακτηριστικά)

ΑΚΡΙΒΕΙΑ	ΚΑΡΡΑ	MAE	RMSE	ROC AUC (ΣΤΑΘΜ.)
89.01%	0.192	0.1868	0.308	0.6

Πίνακας σύγχυσης (γραμμές = πραγματική)

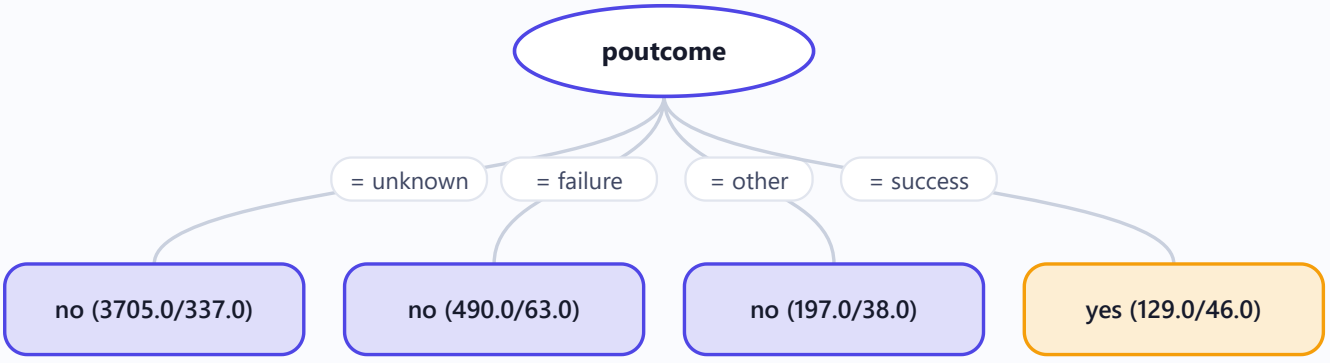
Πραγματική ↓ \ Πρόβλεψη →	no	yes
no	3951	49
yes	448	73

Καμπύλη ROC για «yes» (AUC 0.6)



Κλάση	TP rate	FP rate	Precision	Recall	F-Measure	MCC	ROC	PRC
no	0.988	0.86	0.898	0.988	0.941	0.252	0.6	0.868
yes	0.14	0.012	0.598	0.14	0.227	0.252	0.6	0.249

Δέντρο απόφασης



Μοντέλο (κείμενο)

```
J48 pruned tree
------------------

poutcome = unknown: no (3705.0/337.0)
poutcome = failure: no (490.0/63.0)
poutcome = other: no (197.0/38.0)
poutcome = success: yes (129.0/46.0)

Number of Leaves  :     4

Size of the tree :      5
```

6. NaiveBayes 10-fold cross-validation

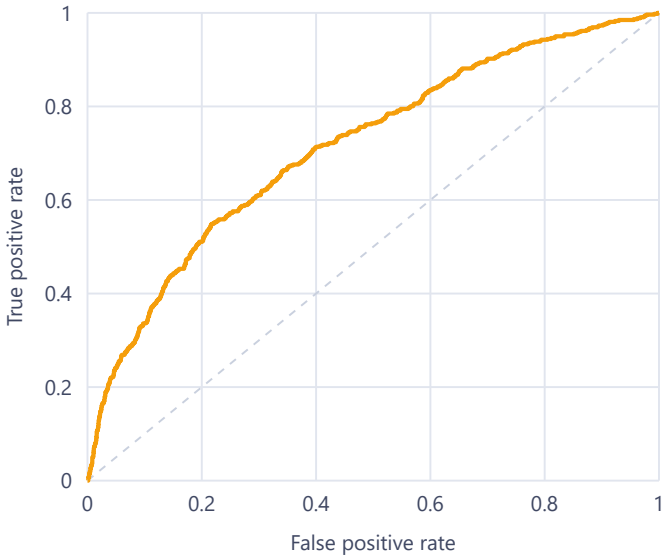
weka.classifiers.bayes.NaiveBayes · bank (16 χαρακτηριστικά)

ΑΚΡΙΒΕΙΑ	ΚΑΡΡΑ	MAE	RMSE	ROC AUC (ΣΤΑΘΜ.)
83.01%	**0.232**	**0.2055**	**0.3617**	**0.715**

Πίνακας σύγχυσης (γραμμές = πραγματική)

Πραγματική ↓ \ Πρόβλεψη →	no	yes
no	3565	435
yes	333	188

Καμπύλη ROC για «yes» (AUC 0.715)



Κλάση	TP rate	FP rate	Precision	Recall	F-Measure	MCC	ROC	PRC
no	0.891	0.639	0.915	0.891	0.903	0.234	0.715	0.942
yes	0.361	0.109	0.302	0.361	0.329	0.234	0.715	0.267

Μοντέλο

```
Naive Bayes Classifier

                      no     yes
Attribute         (0.88)  (0.12)
================================
age
  mean           41.0264 42.5015
  std. dev.      10.2053  13.145
  weight sum        4000     521
  precision       1.0303  1.0303

job
  unemployed         116      14
  services           380      39
  management         839     132
  blue-collar        878      70
  self-employed      164      21
  technician         686      84
  entrepreneur       154      16
  admin.             421      59
  student             66      20
  housemaid           99      15
  retired            177      55
  unknown             32       8
  [total]           4012     533

marital
  married           2521     278
  single            1030     168
  divorced           452      78
  [total]           4003     524

education
  primary            615      65
  secondary         2062     246
  tertiary          1158     194
  unknown            169      20
  [total]           4004     525

default
  no                3934     513
  yes                 68      10
  [total]           4002     523

balance
… (67 ακόμη γραμμές — δες το παράρτημα)
```

7. Logistic

10-fold cross-validation

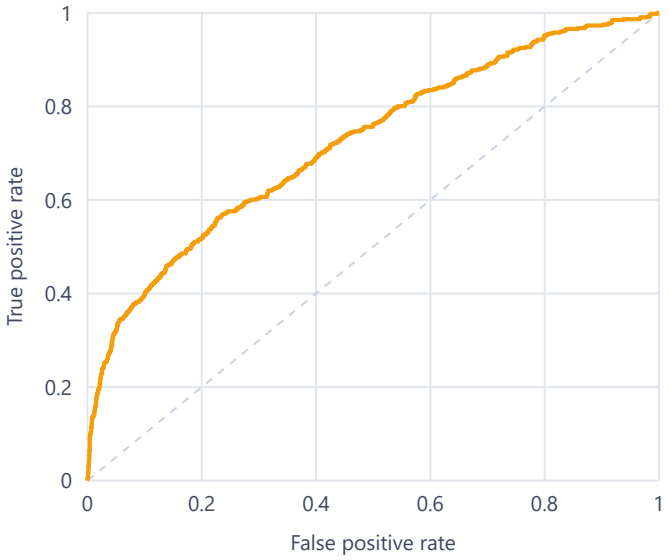
weka.classifiers.functions.Logistic -R 1e-8 -M -1 · bank (16 χαρακτηριστικά)

ΑΚΡΙΒΕΙΑ	ΚΑΡΡΑ	MAE	RMSE	ROC AUC (ΣΤΑΘΜ.)
89.14%	**0.205**	**0.1764**	**0.299**	**0.723**

Πίνακας σύγχυσης (γραμμές = πραγματική)

Πραγματική ↓ \ Πρόβλεψη →	no	yes
no	3952	48
yes	443	78

Καμπύλη ROC για «yes» (AUC 0.723)



Κλάση	TP rate	FP rate	Precision	Recall	F-Measure	MCC	ROC	PRC
no	0.988	0.85	0.899	0.988	0.942	0.267	0.723	0.943
yes	0.15	0.012	0.619	0.15	0.241	0.267	0.723	0.343

Μοντέλο

```
Logistic Regression with ridge parameter of 1.0E-8
Coefficients...
                           Class
Variable                      no
=================================
age                        -0.0017
job=unemployed              0.2349
job=services                0.0723
job=management             -0.0861
job=blue-collar             0.1391
job=self-employed           -0.074
job=technician               0.168
job=entrepreneur           -0.1029
job=admin.                  0.0648
job=student                -0.3437
job=housemaid              -0.0535
job=retired                -0.6843
job=unknown                -0.4421
marital=married             0.2295
marital=single             -0.1036
marital=divorced           -0.3296
education=primary            0.126
education=secondary        -0.0404
education=tertiary         -0.1131
education=unknown           0.4472
default                    -0.3845
balance                          0
housing                     0.1476
loan                        0.5292
contact=cellular            -0.481
contact=unknown             0.6558
contact=telephone          -0.4033
day                        -0.0116
month=oct                  -1.3863
month=may                   0.2051
month=apr                  -0.2963
month=jun                  -0.6954
month=feb                  -0.2957
month=aug                   0.0796
month=jan                   0.7036
month=jul                   0.3338
month=nov                   0.3895
month=sep                  -0.5608
month=mar                  -1.1914
month=dec                  -0.7237
… (62 ακόμη γραμμές — δες το παράρτημα)
```

8. JRip

10-fold cross-validation

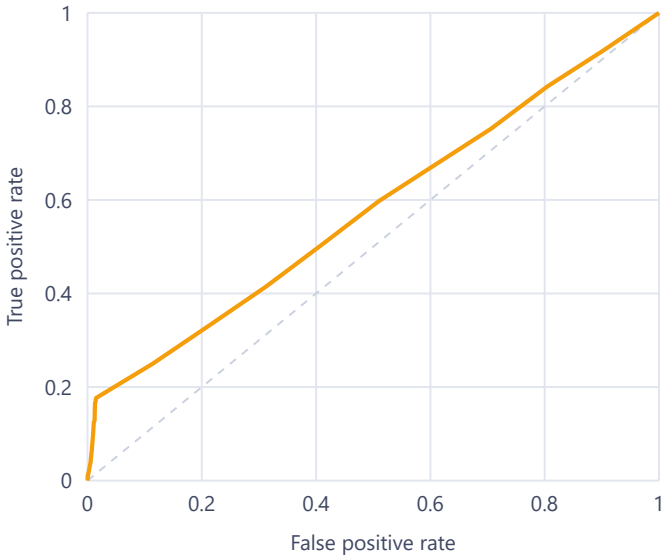
weka.classifiers.rules.JRip -F 3 -N 2 -O 2 -S 1 · bank (16 χαρακτηριστικά)

ΑΚΡΙΒΕΙΑ	ΚΑΡΡΑ	MAE	RMSE	ROC AUC (ΣΤΑΘΜ.)
89.18%	**0.234**	**0.1857**	**0.3064**	**0.579**

Πίνακας σύγχυσης (γραμμές = πραγματική)

Πραγματική ↓ \ Πρόβλεψη →	no	yes
no	3940	60
yes	429	92

Καμπύλη ROC για «yes» (AUC 0.579)



Κλάση	TP rate	FP rate	Precision	Recall	F-Measure	MCC	ROC	PRC
no	0.985	0.823	0.902	0.985	0.942	0.286	0.579	0.907
yes	0.177	0.015	0.605	0.177	0.273	0.286	0.579	0.235

Μοντέλο

```
JRIP rules:
===========

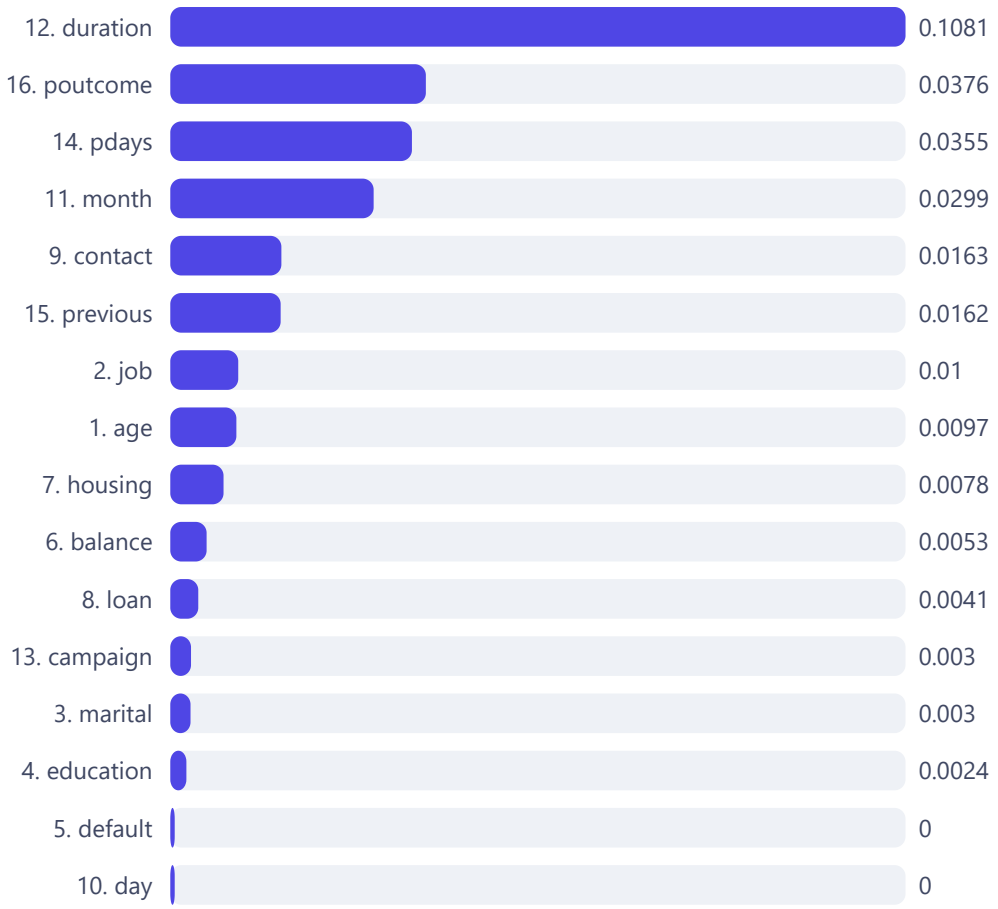
(poutcome = success) => y=yes (129.0/46.0)
(contact = cellular) and (month = oct) and (day >= 20) and (balance <= 944) => y=yes (14.0/2.0)
(contact = cellular) and (month = jun) and (day >= 5) and (balance >= 1315) => y=yes (14.0/3.0)
 => y=no (4364.0/415.0)

Number of Rules : 4
```

5. Επιλογή χαρακτηριστικών

InfoGainAttributeEval + Ranker

weka.attributeSelection.InfoGainAttributeEval · weka.attributeSelection.Ranker -N -1 -T -1.7976931348623157E308 · bank



Επιλεγμένα (16): duration, poutcome, pdays, month, contact, previous, job, age, housing, balance, loan, campaign, marital, education, default, day

InfoGainAttributeEval + Ranker

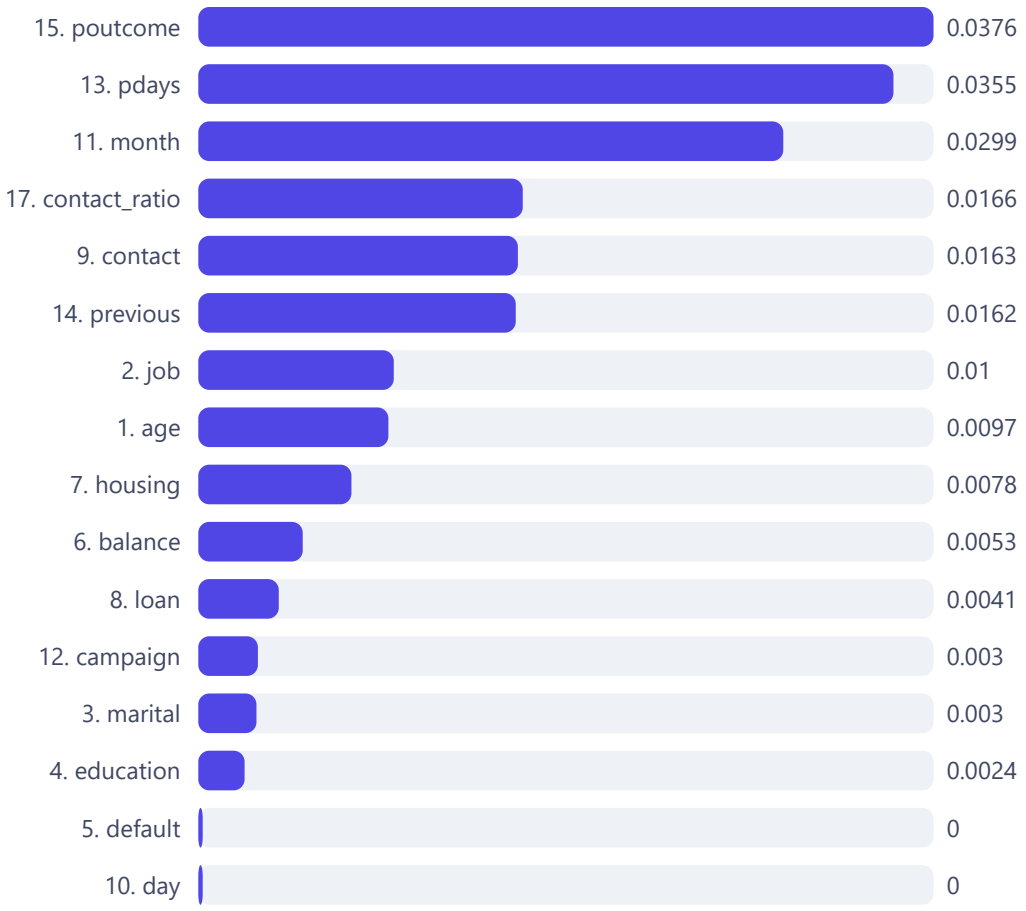
weka.attributeSelection.InfoGainAttributeEval · weka.attributeSelection.Ranker -N -1 -T -1.7976931348623157E308 · bank



Επιλεγμένα (15): poutcome, pdays, month, contact, previous, job, age, housing, balance, loan, campaign, marital, education, default, day

InfoGainAttributeEval + Ranker

weka.attributeSelection.InfoGainAttributeEval · weka.attributeSelection.Ranker -N -1 -T -1.7976931348623157E308 · bank



Επιλεγμένα (16): poutcome, pdays, month, contact_ratio, contact, previous, job, age, housing, balance, loan, campaign, marital, education, default, day

6. Ομαδοποίηση

SimpleKMeans

weka.clusterers.SimpleKMeans -N 4 -init 0 -I 500 -S 10 · bank

Αγνοήθηκαν: contact, day, month, campaign, pdays, previous, poutcome, y

Μέγεθος ομάδων



Κέντρα ομάδων

Χαρακτηριστικό	Όλα	Ομάδα 0	Ομάδα 1	Ομάδα 2	Ομάδα 3
age	41.17	49.96	45.01	41.9	35.8
job	management	self-employed	technician	management	blue-collar
marital	married	divorced	married	married	married
education	secondary	tertiary	secondary	tertiary	secondary
default	no	no	no	no	no
balance	1422.66	1494.46	1387.73	1803.9	1103.73
housing	yes	no	yes	no	yes
loan	no	no	no	no	no

Ομάδες έναντι «y»

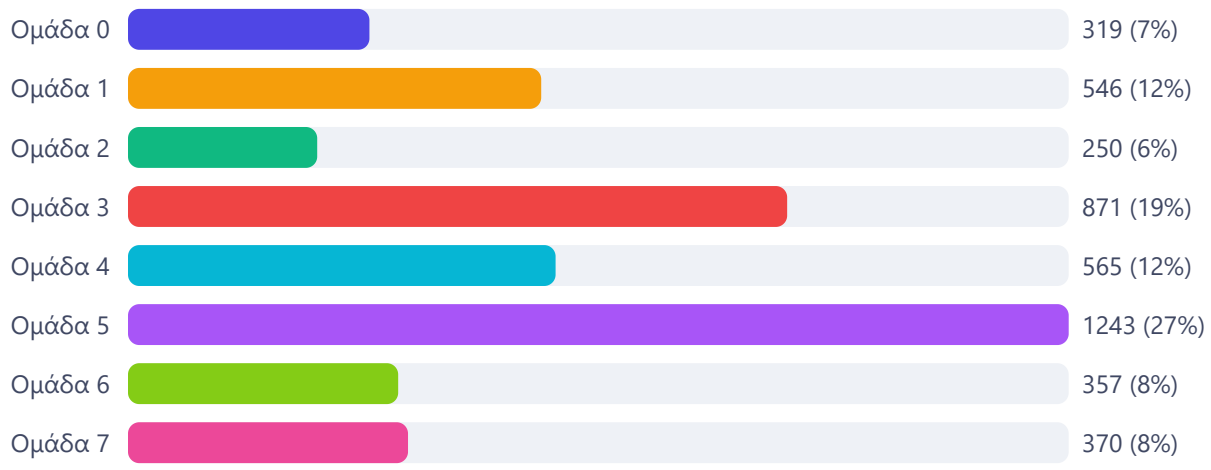
Ομάδα	Μέγεθος	y=no	y=yes	% y=yes
Ομάδα 0	319	257	62	**19.4%**
Ομάδα 1	1231	1094	137	**11.1%**
Ομάδα 2	1382	1195	187	**13.5%**
Ομάδα 3	1589	1454	135	**8.5%**
Σύνολο	4521	4000	521	11.5%

EM

weka.clusterers.EM -N -1 -max -1 -I 100 -M 0.000001 -S 100 · bank

Αγνοήθηκαν: contact, day, month, campaign, pdays, previous, poutcome, y

Μέγεθος ομάδων



Ομάδες έναντι «y»

Ομάδα	Μέγεθος	y=no	y=yes	% y=yes
Ομάδα 0	319	277	42	**13.2%**
Ομάδα 1	546	474	72	**13.2%**
Ομάδα 2	250	182	68	**27.2%**
Ομάδα 3	871	756	115	**13.2%**
Ομάδα 4	565	492	73	**12.9%**
Ομάδα 5	1243	1149	94	**7.6%**
Ομάδα 6	357	328	29	**8.1%**
Ομάδα 7	370	342	28	**7.6%**
Σύνολο	4521	4000	521	11.5%

7. Παράρτημα: πλήρης έξοδος Weka

Classify — J48

```
=== Run information ===

Scheme:       weka.classifiers.trees.J48 -C 0.25 -M 2
Relation:     bank
Instances:    4521
Attributes:   17
              age
              job
              marital
              education
              default
              balance
              housing
              loan
              contact
              day
              month
              duration
              campaign
              pdays
              previous
              poutcome
              y
Test mode:    10-fold cross-validation

=== Classifier model (full training set) ===

J48 pruned tree
------------------

duration <= 211: no (2548.0/73.0)
duration > 211
|   duration <= 645
|   |   poutcome = unknown
|   |   |   age <= 60
|   |   |   |   contact = cellular
|   |   |   |   |   month = oct
|   |   |   |   |   |   balance <= 2469: yes (7.0)
|   |   |   |   |   |   balance > 2469: no (5.0/1.0)
|   |   |   |   |   month = may: no (102.0/14.0)
|   |   |   |   |   month = apr
|   |   |   |   |   |   day <= 20: no (58.0/5.0)
|   |   |   |   |   |   day > 20
|   |   |   |   |   |   |   duration <= 238: no (3.0)
```

```
|   |   |   |   |   |   |   duration > 238: yes (13.0/1.0)
|   |   |   |   |   |   month = jun: yes (21.0/7.0)
|   |   |   |   |   |   month = feb
|   |   |   |   |   |   |   day <= 7
|   |   |   |   |   |   |   |   balance <= 1: yes (3.0/1.0)
|   |   |   |   |   |   |   |   balance > 1: no (37.0)
|   |   |   |   |   |   |   day > 7: yes (9.0/3.0)
|   |   |   |   |   |   month = aug: no (153.0/19.0)
|   |   |   |   |   |   month = jan: no (32.0/1.0)
|   |   |   |   |   |   month = jul: no (192.0/7.0)
|   |   |   |   |   |   month = nov: no (86.0/9.0)
|   |   |   |   |   |   month = sep: no (7.0/1.0)
|   |   |   |   |   |   month = mar
|   |   |   |   |   |   |   housing = no
|   |   |   |   |   |   |   |   duration <= 312: no (2.0)
|   |   |   |   |   |   |   |   duration > 312: yes (2.0)
|   |   |   |   |   |   |   housing = yes: yes (4.0)
|   |   |   |   |   |   month = dec: no (3.0/1.0)
|   |   |   |   |   contact = unknown: no (464.0/16.0)
|   |   |   |   |   contact = telephone: no (50.0/10.0)
|   |   |   |   age > 60
|   |   |   |   |   age <= 68: yes (21.0/8.0)
|   |   |   |   |   age > 68: no (16.0/7.0)
|   |   |   poutcome = failure
|   |   |   |   pdays <= 373: no (163.0/26.0)
|   |   |   |   pdays > 373
|   |   |   |   |   balance <= 2581: yes (9.0)
|   |   |   |   |   balance > 2581: no (2.0)
|   |   |   poutcome = other
|   |   |   |   month = oct: yes (3.0/1.0)
|   |   |   |   month = may: no (20.0/1.0)
|   |   |   |   month = apr
|   |   |   |   |   campaign <= 5: no (11.0/1.0)
|   |   |   |   |   campaign > 5: yes (2.0)
|   |   |   |   month = jun: yes (7.0/2.0)
|   |   |   |   month = feb: no (4.0/1.0)
|   |   |   |   month = aug: yes (6.0/1.0)
|   |   |   |   month = jan: no (6.0)
|   |   |   |   month = jul: no (1.0)
|   |   |   |   month = nov
|   |   |   |   |   age <= 32: yes (4.0)
|   |   |   |   |   age > 32: no (6.0)
|   |   |   |   month = sep: no (3.0/1.0)
|   |   |   |   month = mar: no (1.0)
|   |   |   |   month = dec: no (0.0)
|   |   |   poutcome = success: yes (76.0/16.0)
|   |   duration > 645
|   |   |   marital = married
|   |   |   |   default = no
|   |   |   |   |   contact = cellular
|   |   |   |   |   |   job = unemployed
|   |   |   |   |   |   |   balance <= 640: no (3.0)
|   |   |   |   |   |   |   balance > 640: yes (2.0)
|   |   |   |   |   |   job = services
|   |   |   |   |   |   |   loan = no: yes (6.0)
|   |   |   |   |   |   |   loan = yes: no (4.0/1.0)
|   |   |   |   |   |   job = management
|   |   |   |   |   |   |   poutcome = unknown
|   |   |   |   |   |   |   |   month = oct: no (0.0)
|   |   |   |   |   |   |   |   month = may: yes (3.0/1.0)
|   |   |   |   |   |   |   |   month = apr: no (1.0)
|   |   |   |   |   |   |   |   month = jun: no (1.0)
|   |   |   |   |   |   |   |   month = feb: yes (1.0)
|   |   |   |   |   |   |   |   month = aug: yes (5.0/1.0)
|   |   |   |   |   |   |   |   month = jan: no (1.0)
|   |   |   |   |   |   |   |   month = jul
|   |   |   |   |   |   |   |   |   age <= 43: no (8.0/2.0)
|   |   |   |   |   |   |   |   |   age > 43: yes (2.0)
|   |   |   |   |   |   |   |   month = nov
|   |   |   |   |   |   |   |   |   age <= 50: yes (4.0/1.0)
|   |   |   |   |   |   |   |   |   age > 50: no (3.0)
|   |   |   |   |   |   |   |   month = sep: no (0.0)
|   |   |   |   |   |   |   |   month = mar: no (0.0)
|   |   |   |   |   |   |   |   month = dec: no (0.0)
|   |   |   |   |   |   |   poutcome = failure
|   |   |   |   |   |   |   |   education = primary: no (0.0)
```

```
|   |   |   |   |   |   |   education = secondary: yes (2.0)
|   |   |   |   |   |   |   education = tertiary: no (3.0)
|   |   |   |   |   |   |   education = unknown: no (0.0)
|   |   |   |   |   |   poutcome = other: yes (2.0)
|   |   |   |   |   |   poutcome = success: yes (1.0)
|   |   |   |   |   job = blue-collar
|   |   |   |   |   |   housing = no
|   |   |   |   |   |   |   campaign <= 3: no (7.0)
|   |   |   |   |   |   |   campaign > 3: yes (2.0)
|   |   |   |   |   |   housing = yes
|   |   |   |   |   |   |   previous <= 1
|   |   |   |   |   |   |   |   campaign <= 1: no (7.0/1.0)
|   |   |   |   |   |   |   |   campaign > 1
|   |   |   |   |   |   |   |   |   duration <= 1073: yes (4.0)
|   |   |   |   |   |   |   |   |   duration > 1073: no (2.0)
|   |   |   |   |   |   |   previous > 1: yes (3.0)
|   |   |   |   |   job = self-employed
|   |   |   |   |   |   campaign <= 3: no (2.0)
|   |   |   |   |   |   campaign > 3: yes (3.0)
|   |   |   |   |   job = technician: yes (10.0/2.0)
|   |   |   |   |   job = entrepreneur: no (7.0/2.0)
|   |   |   |   |   job = admin.: no (5.0/1.0)
|   |   |   |   |   job = student: no (2.0/1.0)
|   |   |   |   |   job = housemaid: no (2.0)
|   |   |   |   |   job = retired
|   |   |   |   |   |   pdays <= 28: no (3.0)
|   |   |   |   |   |   pdays > 28: yes (2.0)
|   |   |   |   |   job = unknown: yes (3.0/1.0)
|   |   |   |   contact = unknown: no (67.0/18.0)
|   |   |   |   contact = telephone: no (19.0/7.0)
|   |   |   default = yes: yes (3.0)
|   |   marital = single: yes (101.0/39.0)
|   |   marital = divorced
|   |   |   poutcome = unknown
|   |   |   |   duration <= 924
|   |   |   |   |   job = unemployed: no (2.0)
|   |   |   |   |   job = services: no (2.0/1.0)
|   |   |   |   |   job = management: no (4.0/1.0)
|   |   |   |   |   job = blue-collar
|   |   |   |   |   |   balance <= -145: yes (2.0)
|   |   |   |   |   |   balance > -145: no (5.0)
|   |   |   |   |   job = self-employed: yes (2.0)
|   |   |   |   |   job = technician: no (0.0)
|   |   |   |   |   job = entrepreneur: no (2.0)
|   |   |   |   |   job = admin.: yes (4.0/1.0)
|   |   |   |   |   job = student: no (0.0)
|   |   |   |   |   job = housemaid: yes (1.0)
|   |   |   |   |   job = retired: yes (2.0)
|   |   |   |   |   job = unknown: no (0.0)
|   |   |   |   duration > 924: yes (18.0/1.0)
|   |   |   poutcome = failure
|   |   |   |   duration <= 834: yes (2.0)
|   |   |   |   duration > 834: no (3.0)
|   |   |   poutcome = other: yes (3.0/1.0)
|   |   |   poutcome = success: no (1.0)

Number of Leaves  :     104

Size of the tree :      146


Time taken to build model: 0.11 seconds

=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances        4023               88.9847 %
Incorrectly Classified Instances       498               11.0153 %
Kappa statistic                          0.368
Mean absolute error                      0.1448
Root mean squared error                  0.2977
Relative absolute error                 70.9698 %
Root relative squared error             93.2371 %
Total Number of Instances             4521

=== Detailed Accuracy By Class ===
```

```
                TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Class
                0.960    0.645    0.920      0.960    0.939      0.377    0.762     0.932     no
                0.355    0.041    0.533      0.355    0.426      0.377    0.762     0.405     yes
Weighted Avg.   0.890    0.575    0.875      0.890    0.880      0.377    0.762     0.871

=== Confusion Matrix ===

    a    b   <-- classified as
 3838  162 | a = no
  336  185 | b = yes
```

Classify — NaiveBayes

```
=== Run information ===

Scheme:       weka.classifiers.bayes.NaiveBayes
Relation:     bank
Instances:    4521
Attributes:   17
              age
              job
              marital
              education
              default
              balance
              housing
              loan
              contact
              day
              month
              duration
              campaign
              pdays
              previous
              poutcome
              y
Test mode:    10-fold cross-validation

=== Classifier model (full training set) ===

Naive Bayes Classifier

                     no     yes
Attribute         (0.88)  (0.12)
================================
age
  mean           41.0264 42.5015
  std. dev.      10.2053  13.145
  weight sum        4000     521
  precision       1.0303  1.0303

job
  unemployed         116      14
  services           380      39
  management         839     132
  blue-collar        878      70
  self-employed      164      21
  technician         686      84
  entrepreneur       154      16
  admin.             421      59
  student             66      20
  housemaid           99      15
  retired            177      55
  unknown             32       8
  [total]           4012     533

marital
  married           2521     278
  single            1030     168
  divorced           452      78
  [total]           4003     524
```

```
education
  primary            615      65
  secondary         2062     246
  tertiary          1158     194
  unknown            169      20
  [total]           4004     525

default
  no                3934     513
  yes                 68      10
  [total]           4002     523

balance
  mean          1403.0309 1572.1066
  std. dev.     3075.0271 2442.0069
  weight sum        4000     521
  precision      31.6756  31.6756

housing
  no                1662     302
  yes               2340     221
  [total]           4002     523

loan
  no                3353     479
  yes                649      44
  [total]           4002     523

contact
  cellular          2481     417
  unknown           1264      62
  telephone          258      45
  [total]           4003     524

day
  mean           15.9488  15.6583
  std. dev.       8.2487   8.2272
  weight sum        4000     521
  precision            1        1

month
  oct                 44      38
  may               1306      94
  apr                238      57
  jun                477      56
  feb                185      39
  aug                555      80
  jan                133      17
  jul                646      62
  nov                351      40
  sep                 36      18
  mar                 29      22
  dec                 12      10
  [total]           4012     533

duration
  mean          226.3607 552.8113
  std. dev.      210.2943 89.9421
  weight sum        4000     521
  precision       3.4565   3.4565

campaign
  mean             3.023   2.4999
  std. dev.       3.1626   2.0411
  weight sum        4000     521
  precision       1.5806   1.5806

pdays
  mean           36.8015   69.243
  std. dev.      95.8516 121.4075
  weight sum        4000     521
  precision       2.9966   2.9966

previous
  mean            0.4943   1.1558
```

```
  std. dev.       1.6518  2.1173
  weight sum        4000     521
  precision        1.087   1.087

poutcome
  unknown           3369     338
  failure            428      64
  other              160      39
  success             47      84
  [total]           4004     525


Time taken to build model: 0.03 seconds

=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances        3928               86.8834 %
Incorrectly Classified Instances       593               13.1166 %
Kappa statistic                          0.3975
Mean absolute error                      0.1625
Root mean squared error                  0.3233
Relative absolute error                 79.6454 %
Root relative squared error            101.2447 %
Total Number of Instances             4521

=== Detailed Accuracy By Class ===

                 TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Class
                 0.916    0.491    0.935      0.916    0.925      0.399    0.845     0.972     no
                 0.509    0.084    0.440      0.509    0.472      0.399    0.845     0.405     yes
Weighted Avg.    0.869    0.444    0.878      0.869    0.873      0.399    0.845     0.907

=== Confusion Matrix ===

    a    b   <-- classified as
 3663  337 | a = no
  256  265 | b = yes
```

Classify — Logistic

```
=== Run information ===

Scheme:       weka.classifiers.functions.Logistic -R 1e-8 -M -1
Relation:     bank
Instances:    4521
Attributes:   17
              age
              job
              marital
              education
              default
              balance
              housing
              loan
              contact
              day
              month
              duration
              campaign
              pdays
              previous
              poutcome
              y
Test mode:    10-fold cross-validation

=== Classifier model (full training set) ===

Logistic Regression with ridge parameter of 1.0E-8
Coefficients...
                           Class
Variable                      no
```

```
==================================
age                           0.0042
job=unemployed                0.5026
job=services                  0.0088
job=management               -0.0639
job=blue-collar               0.2555
job=self-employed             0.0443
job=technician                0.0557
job=entrepreneur              0.1129
job=admin.                   -0.1369
job=student                  -0.5153
job=housemaid                 0.2161
job=retired                  -0.7684
job=unknown                  -0.6576
marital=married               0.1507
marital=single               -0.0137
marital=divorced             -0.3188
education=primary             0.1125
education=secondary           0.0324
education=tertiary           -0.2082
education=unknown             0.5336
default                      -0.5446
balance                            0
housing                         0.26
loan                          0.6296
contact=cellular             -0.5959
contact=unknown               0.8202
contact=telephone            -0.5257
day                          -0.0164
month=oct                    -1.6355
month=may                     0.2155
month=apr                    -0.2745
month=jun                    -0.8287
month=feb                    -0.4767
month=aug                     0.0337
month=jan                     0.8488
month=jul                      0.477
month=nov                     0.5685
month=sep                    -0.9317
month=mar                     -1.773
month=dec                    -0.3889
duration                     -0.0042
campaign                      0.0704
pdays                         0.0001
previous                      0.0055
poutcome=unknown              0.3454
poutcome=failure              0.2238
poutcome=other               -0.2674
poutcome=success             -2.2212
Intercept                     3.4516


Odds Ratios...
                               Class
Variable                          no
==================================
age                           1.0042
job=unemployed                1.6531
job=services                  1.0088
job=management                0.9381
job=blue-collar               1.2911
job=self-employed             1.0453
job=technician                1.0573
job=entrepreneur              1.1195
job=admin.                    0.8721
job=student                   0.5973
job=housemaid                 1.2412
job=retired                   0.4638
job=unknown                   0.5181
marital=married               1.1627
marital=single                0.9864
marital=divorced               0.727
education=primary             1.1191
education=secondary            1.033
education=tertiary             0.812
education=unknown              1.705
```

```
default                      0.58
balance                         1
housing                    1.2969
loan                       1.8769
contact=cellular           0.5511
contact=unknown             2.271
contact=telephone          0.5912
day                        0.9837
month=oct                  0.1949
month=may                  1.2405
month=apr                    0.76
month=jun                  0.4366
month=feb                  0.6209
month=aug                  1.0342
month=jan                  2.3369
month=jul                  1.6112
month=nov                  1.7656
month=sep                  0.3939
month=mar                  0.1698
month=dec                  0.6778
duration                   0.9958
campaign                    1.073
pdays                      1.0001
previous                   1.0055
poutcome=unknown           1.4125
poutcome=failure           1.2508
poutcome=other             0.7654
poutcome=success           0.1085


Time taken to build model: 0.08 seconds

=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances        4077               90.1792 %
Incorrectly Classified Instances       444                9.8208 %
Kappa statistic                          0.3862
Mean absolute error                      0.1422
Root mean squared error                  0.2723
Relative absolute error                 69.6955 %
Root relative squared error             85.2822 %
Total Number of Instances             4521

=== Detailed Accuracy By Class ===

                 TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Class
                 0.977    0.674    0.918      0.977    0.946      0.413    0.888     0.983     no
                 0.326    0.023    0.646      0.326    0.434      0.413    0.888     0.506     yes
Weighted Avg.    0.902    0.599    0.886      0.902    0.887      0.413    0.888     0.928

=== Confusion Matrix ===

    a    b   <-- classified as
 3907   93 | a = no
  351  170 | b = yes
```

Classify — JRip

```
=== Run information ===

Scheme:       weka.classifiers.rules.JRip -F 3 -N 2 -O 2 -S 1
Relation:     bank
Instances:    4521
Attributes:   17
              age
              job
              marital
              education
              default
              balance
              housing
```

```
              loan
              contact
              day
              month
              duration
              campaign
              pdays
              previous
              poutcome
              y
Test mode:    10-fold cross-validation

=== Classifier model (full training set) ===

JRIP rules:
===========

(duration >= 651) and (contact = cellular) and (campaign >= 4) and (loan = no) => y=yes (40.0/10.0)
(duration >= 757) => y=yes (207.0/98.0)
(duration >= 212) and (pdays >= 56) and (poutcome = success) => y=yes (82.0/19.0)
(duration >= 212) and (contact = cellular) and (pdays >= 375) => y=yes (11.0/1.0)
(duration >= 342) and (housing = no) and (month = oct) => y=yes (14.0/3.0)
 => y=no (4167.0/298.0)

Number of Rules : 6


Time taken to build model: 0.31 seconds

=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances        4057               89.7368 %
Incorrectly Classified Instances       464               10.2632 %
Kappa statistic                          0.3873
Mean absolute error                      0.1651
Root mean squared error                  0.2916
Relative absolute error                 80.9244 %
Root relative squared error             91.3167 %
Total Number of Instances             4521

=== Detailed Accuracy By Class ===

                 TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Class
                 0.969    0.651    0.920      0.969    0.944      0.404    0.670     0.925     no
                 0.349    0.031    0.593      0.349    0.440      0.404    0.670     0.358     yes
Weighted Avg.    0.897    0.579    0.882      0.897    0.885      0.404    0.670     0.859

=== Confusion Matrix ===

    a    b   <-- classified as
 3875  125 | a = no
  339  182 | b = yes
```

Classify — J48

```
=== Run information ===

Scheme:       weka.classifiers.trees.J48 -C 0.25 -M 2
Relation:     bank
Instances:    4521
Attributes:   16
              age
              job
              marital
              education
              default
              balance
              housing
              loan
              contact
              day
```

```
              month
              campaign
              pdays
              previous
              poutcome
              y
Test mode:    10-fold cross-validation

=== Classifier model (full training set) ===

J48 pruned tree
------------------

poutcome = unknown: no (3705.0/337.0)
poutcome = failure: no (490.0/63.0)
poutcome = other: no (197.0/38.0)
poutcome = success: yes (129.0/46.0)

Number of Leaves  :     4

Size of the tree :      5


Time taken to build model: 0.09 seconds

=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances        4024               89.0069 %
Incorrectly Classified Instances       497               10.9931 %
Kappa statistic                          0.1917
Mean absolute error                      0.1868
Root mean squared error                  0.308
Relative absolute error                 91.5201 %
Root relative squared error             96.4603 %
Total Number of Instances             4521

=== Detailed Accuracy By Class ===

                 TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Class
                 0.988    0.860    0.898      0.988    0.941      0.252    0.600     0.868     no
                 0.140    0.012    0.598      0.140    0.227      0.252    0.600     0.249     yes
Weighted Avg.    0.890    0.762    0.864      0.890    0.859      0.252    0.600     0.796

=== Confusion Matrix ===

    a    b   <-- classified as
 3951   49 | a = no
  448   73 | b = yes
```

Classify — NaiveBayes

```
=== Run information ===

Scheme:       weka.classifiers.bayes.NaiveBayes
Relation:     bank
Instances:    4521
Attributes:   16
              age
              job
              marital
              education
              default
              balance
              housing
              loan
              contact
              day
              month
              campaign
              pdays
              previous
```

```
              poutcome
              y
Test mode:    10-fold cross-validation

=== Classifier model (full training set) ===

Naive Bayes Classifier

                       no     yes
Attribute           (0.88)  (0.12)
================================
age
  mean             41.0264 42.5015
  std. dev.        10.2053  13.145
  weight sum          4000     521
  precision         1.0303  1.0303

job
  unemployed           116      14
  services             380      39
  management           839     132
  blue-collar          878      70
  self-employed        164      21
  technician           686      84
  entrepreneur         154      16
  admin.               421      59
  student               66      20
  housemaid             99      15
  retired              177      55
  unknown               32       8
  [total]             4012     533

marital
  married             2521     278
  single              1030     168
  divorced             452      78
  [total]             4003     524

education
  primary              615      65
  secondary           2062     246
  tertiary            1158     194
  unknown              169      20
  [total]             4004     525

default
  no                  3934     513
  yes                   68      10
  [total]             4002     523

balance
  mean           1403.03091572.1066
  std. dev.      3075.02712442.0069
  weight sum          4000     521
  precision       31.6756 31.6756

housing
  no                  1662     302
  yes                 2340     221
  [total]             4002     523

loan
  no                  3353     479
  yes                  649      44
  [total]             4002     523

contact
  cellular            2481     417
  unknown             1264      62
  telephone            258      45
  [total]             4003     524

day
  mean             15.9488 15.6583
  std. dev.         8.2487  8.2272
  weight sum          4000     521
```

```
  precision            1       1

month
  oct                 44      38
  may               1306      94
  apr                238      57
  jun                477      56
  feb                185      39
  aug                555      80
  jan                133      17
  jul                646      62
  nov                351      40
  sep                 36      18
  mar                 29      22
  dec                 12      10
  [total]           4012     533

campaign
  mean             3.023  2.4999
  std. dev.       3.1626  2.0411
  weight sum        4000     521
  precision       1.5806  1.5806

pdays
  mean           36.8015  69.243
  std. dev.      95.8516121.4075
  weight sum        4000     521
  precision       2.9966  2.9966

previous
  mean             0.4943  1.1558
  std. dev.        1.6518  2.1173
  weight sum         4000     521
  precision         1.087   1.087

poutcome
  unknown           3369     338
  failure            428      64
  other              160      39
  success             47      84
  [total]           4004     525


Time taken to build model: 0.03 seconds

=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances        3753               83.0126 %
Incorrectly Classified Instances       768               16.9874 %
Kappa statistic                          0.2323
Mean absolute error                      0.2055
Root mean squared error                  0.3617
Relative absolute error                100.6871 %
Root relative squared error            113.2725 %
Total Number of Instances             4521

=== Detailed Accuracy By Class ===

                 TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Class
                 0.891    0.639    0.915      0.891    0.903      0.234    0.715     0.942     no
                 0.361    0.109    0.302      0.361    0.329      0.234    0.715     0.267     yes
Weighted Avg.    0.830    0.578    0.844      0.830    0.837      0.234    0.715     0.864

=== Confusion Matrix ===

    a    b   <-- classified as
 3565  435 | a = no
  333  188 | b = yes
```

Classify — Logistic

```
=== Run information ===

Scheme:       weka.classifiers.functions.Logistic -R 1e-8 -M -1
Relation:     bank
Instances:    4521
Attributes:   16
              age
              job
              marital
              education
              default
              balance
              housing
              loan
              contact
              day
              month
              campaign
              pdays
              previous
              poutcome
              y
Test mode:    10-fold cross-validation

=== Classifier model (full training set) ===

Logistic Regression with ridge parameter of 1.0E-8
Coefficients...
                           Class
Variable                      no
================================
age                       -0.0017
job=unemployed             0.2349
job=services               0.0723
job=management            -0.0861
job=blue-collar            0.1391
job=self-employed          -0.074
job=technician              0.168
job=entrepreneur          -0.1029
job=admin.                 0.0648
job=student               -0.3437
job=housemaid             -0.0535
job=retired               -0.6843
job=unknown               -0.4421
marital=married            0.2295
marital=single            -0.1036
marital=divorced          -0.3296
education=primary           0.126
education=secondary       -0.0404
education=tertiary        -0.1131
education=unknown          0.4472
default                   -0.3845
balance                         0
housing                    0.1476
loan                       0.5292
contact=cellular           -0.481
contact=unknown            0.6558
contact=telephone         -0.4033
day                       -0.0116
month=oct                 -1.3863
month=may                  0.2051
month=apr                 -0.2963
month=jun                 -0.6954
month=feb                 -0.2957
month=aug                  0.0796
month=jan                  0.7036
month=jul                  0.3338
month=nov                  0.3895
month=sep                 -0.5608
month=mar                 -1.1914
month=dec                 -0.7237
```

```
campaign                  0.0669
pdays                    -0.0001
previous                 -0.0106
poutcome=unknown          0.2662
poutcome=failure          0.3005
poutcome=other           -0.2362
poutcome=success         -2.1129
Intercept                 2.1374


Odds Ratios...
                           Class
Variable                      no
===================================
age                       0.9983
job=unemployed            1.2648
job=services               1.075
job=management            0.9175
job=blue-collar           1.1492
job=self-employed         0.9287
job=technician            1.1829
job=entrepreneur          0.9022
job=admin.                1.0669
job=student               0.7091
job=housemaid             0.9479
job=retired               0.5045
job=unknown               0.6427
marital=married            1.258
marital=single            0.9016
marital=divorced          0.7192
education=primary         1.1342
education=secondary       0.9604
education=tertiary        0.8931
education=unknown          1.564
default                   0.6808
balance                        1
housing                   1.1591
loan                      1.6975
contact=cellular          0.6182
contact=unknown           1.9267
contact=telephone         0.6681
day                       0.9885
month=oct                   0.25
month=may                 1.2276
month=apr                 0.7436
month=jun                 0.4989
month=feb                  0.744
month=aug                 1.0828
month=jan                 2.0211
month=jul                 1.3963
month=nov                 1.4762
month=sep                 0.5708
month=mar                 0.3038
month=dec                  0.485
campaign                  1.0692
pdays                     0.9999
previous                  0.9895
poutcome=unknown           1.305
poutcome=failure          1.3506
poutcome=other            0.7896
poutcome=success          0.1209


Time taken to build model: 0.08 seconds

=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances        4030               89.1396 %
Incorrectly Classified Instances       491               10.8604 %
Kappa statistic                          0.2054
Mean absolute error                      0.1764
Root mean squared error                  0.299
Relative absolute error                 86.4626 %
Root relative squared error             93.6341 %
Total Number of Instances             4521
```

```
=== Detailed Accuracy By Class ===

                 TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Class
                 0.988    0.850    0.899      0.988    0.942      0.267    0.723     0.943     no
                 0.150    0.012    0.619      0.150    0.241      0.267    0.723     0.343     yes
Weighted Avg.    0.891    0.754    0.867      0.891    0.861      0.267    0.723     0.874

=== Confusion Matrix ===

    a    b   <-- classified as
 3952   48 | a = no
  443   78 | b = yes
```

Classify — JRip

```
=== Run information ===

Scheme:       weka.classifiers.rules.JRip -F 3 -N 2 -O 2 -S 1
Relation:     bank
Instances:    4521
Attributes:   16
              age
              job
              marital
              education
              default
              balance
              housing
              loan
              contact
              day
              month
              campaign
              pdays
              previous
              poutcome
              y
Test mode:    10-fold cross-validation

=== Classifier model (full training set) ===

JRIP rules:
===========

(poutcome = success) => y=yes (129.0/46.0)
(contact = cellular) and (month = oct) and (day >= 20) and (balance <= 944) => y=yes (14.0/2.0)
(contact = cellular) and (month = jun) and (day >= 5) and (balance >= 1315) => y=yes (14.0/3.0)
 => y=no (4364.0/415.0)

Number of Rules : 4


Time taken to build model: 0.25 seconds

=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances        4032               89.1838 %
Incorrectly Classified Instances       489               10.8162 %
Kappa statistic                          0.2335
Mean absolute error                      0.1857
Root mean squared error                  0.3064
Relative absolute error                 91.0006 %
Root relative squared error             95.9591 %
Total Number of Instances             4521

=== Detailed Accuracy By Class ===

                 TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Class
                 0.985    0.823    0.902      0.985    0.942      0.286    0.579     0.907     no
                 0.177    0.015    0.605      0.177    0.273      0.286    0.579     0.235     yes
```

```
Weighted Avg.    0.892    0.730    0.868      0.892    0.865      0.286    0.579     0.829

=== Confusion Matrix ===

    a    b   <-- classified as
 3940   60 | a = no
  429   92 | b = yes
```

Select attributes — InfoGainAttributeEval + Ranker

```
=== Run information ===

Evaluator:    weka.attributeSelection.InfoGainAttributeEval
Search:       weka.attributeSelection.Ranker -N -1 -T -1.7976931348623157E308
Relation:     bank
Instances:    4521
Attributes:   17
              age
              job
              marital
              education
              default
              balance
              housing
              loan
              contact
              day
              month
              duration
              campaign
              pdays
              previous
              poutcome
              y
Evaluation mode:    evaluate on all training data


=== Attribute Selection on all input data ===

Search Method:
        Attribute ranking.

Attribute Evaluator (supervised, Class (nominal): 17 y):
        Information Gain Ranking Filter

Ranked attributes:
    0.1081  12 duration
    0.0376  16 poutcome
    0.0355  14 pdays
    0.0299  11 month
    0.0163   9 contact
    0.0162  15 previous
    0.01     2 job
    0.0097   1 age
    0.0078   7 housing
    0.0053   6 balance
    0.0041   8 loan
    0.003   13 campaign
    0.003    3 marital
    0.0024   4 education
    0        5 default
    0       10 day

Selected attributes: 12,16,14,11,9,15,2,1,7,6,8,13,3,4,5,10 : 16
                     duration
                     poutcome
                     pdays
                     month
                     contact
                     previous
                     job
                     age
```

```
                     housing
                     balance
                     loan
                     campaign
                     marital
                     education
                     default
                     day
```

Select attributes — InfoGainAttributeEval + Ranker

```
=== Run information ===

Evaluator:    weka.attributeSelection.InfoGainAttributeEval
Search:       weka.attributeSelection.Ranker -N -1 -T -1.7976931348623157E308
Relation:     bank
Instances:    4521
Attributes:   16
              age
              job
              marital
              education
              default
              balance
              housing
              loan
              contact
              day
              month
              campaign
              pdays
              previous
              poutcome
              y
Evaluation mode:    evaluate on all training data


=== Attribute Selection on all input data ===

Search Method:
        Attribute ranking.

Attribute Evaluator (supervised, Class (nominal): 16 y):
        Information Gain Ranking Filter

Ranked attributes:
    0.0376  15 poutcome
    0.0355  13 pdays
    0.0299  11 month
    0.0163   9 contact
    0.0162  14 previous
    0.01     2 job
    0.0097   1 age
    0.0078   7 housing
    0.0053   6 balance
    0.0041   8 loan
    0.003   12 campaign
    0.003    3 marital
    0.0024   4 education
    0        5 default
    0       10 day

Selected attributes: 15,13,11,9,14,2,1,7,6,8,12,3,4,5,10 : 15
                     poutcome
                     pdays
                     month
                     contact
                     previous
                     job
                     age
                     housing
                     balance
```

```
                     loan
                     campaign
                     marital
                     education
                     default
                     day
```

Select attributes — InfoGainAttributeEval + Ranker

```
=== Run information ===

Evaluator:    weka.attributeSelection.InfoGainAttributeEval
Search:       weka.attributeSelection.Ranker -N -1 -T -1.7976931348623157E308
Relation:     bank
Instances:    4521
Attributes:   17
              age
              job
              marital
              education
              default
              balance
              housing
              loan
              contact
              day
              month
              campaign
              pdays
              previous
              poutcome
              y
              contact_ratio
Evaluation mode:    evaluate on all training data


=== Attribute Selection on all input data ===

Search Method:
        Attribute ranking.

Attribute Evaluator (supervised, Class (nominal): 16 y):
        Information Gain Ranking Filter

Ranked attributes:
    0.0376  15 poutcome
    0.0355  13 pdays
    0.0299  11 month
    0.0166  17 contact_ratio
    0.0163   9 contact
    0.0162  14 previous
    0.01     2 job
    0.0097   1 age
    0.0078   7 housing
    0.0053   6 balance
    0.0041   8 loan
    0.003   12 campaign
    0.003    3 marital
    0.0024   4 education
    0        5 default
    0       10 day

Selected attributes: 15,13,11,17,9,14,2,1,7,6,8,12,3,4,5,10 : 16
                     poutcome
                     pdays
                     month
                     contact_ratio
                     contact
                     previous
                     job
                     age
                     housing
```

```
                    balance
                    loan
                    campaign
                    marital
                    education
                    default
                    day
```

Cluster — SimpleKMeans

```
=== Run information ===

Scheme:       weka.clusterers.SimpleKMeans -N 4 -init 0 -I 500 -S 10
Relation:     bank
Instances:    4521
Attributes:   16
              age
              job
              marital
              education
              default
              balance
              housing
              loan
              contact   [αγνοείται]
              day   [αγνοείται]
              month   [αγνοείται]
              campaign   [αγνοείται]
              pdays   [αγνοείται]
              previous   [αγνοείται]
              poutcome   [αγνοείται]
              y   [αγνοείται]
Test mode:    evaluate on training data

=== Clustering model (full training set) ===


kMeans
======

Number of iterations: 5
Within cluster sum of squared errors: 6989.338341635454

Initial starting points (random):

Cluster 0: 61.0,self-employed,divorced,tertiary,no,6610.0,no,no
Cluster 1: 45.0,technician,married,secondary,no,999.0,yes,no
Cluster 2: 37.0,management,married,tertiary,no,1315.0,no,no
Cluster 3: 44.0,management,married,secondary,no,-1746.0,yes,no

Missing values globally replaced with mean/mode

Final cluster centroids:
                                                                  Cluster#
Attribute      Full Data             0             1             2             3
                (4521.0)       (319.0)      (1231.0)      (1382.0)      (1589.0)
================================================================================
age               41.1701       49.9592       45.0073       41.8965       35.8011
job           management self-employed    technician    management   blue-collar
marital           married      divorced       married       married       married
education       secondary      tertiary     secondary      tertiary     secondary
default                no            no            no            no            no
balance        1422.6578     1494.4577     1387.7254     1803.9023     1103.7262
housing              yes            no           yes            no           yes
loan                  no            no            no            no            no


Time taken to build model (full training data) : 0.03 seconds

=== Model and evaluation on training set ===

Clustered Instances
```

```
0      319 (  7%)
1      1231 ( 27%)
2      1382 ( 31%)
3      1589 ( 35%)
```

Cluster — EM

```
=== Run information ===

Scheme:       weka.clusterers.EM -N -1 -max -1 -I 100 -M 0.000001 -S 100
Relation:     bank
Instances:    4521
Attributes:   16
              age
              job
              marital
              education
              default
              balance
              housing
              loan
              contact   [αγνοείται]
              day   [αγνοείται]
              month   [αγνοείται]
              campaign   [αγνοείται]
              pdays   [αγνοείται]
              previous   [αγνοείται]
              poutcome   [αγνοείται]
              y   [αγνοείται]
Test mode:    evaluate on training data

=== Clustering model (full training set) ===

EM
==

Number of clusters selected by cross validation: 8
Number of iterations performed: 100

                        Cluster
Attribute                    0            1            2            3            4            5            6
7
                        (0.08)       (0.14)       (0.06)       (0.18)       (0.13)       (0.25)       (0.08)
(0.08)
=============================================================================================================================
=======
age
  mean                41.5594      39.5354      60.4805      38.8063       30.799      40.3782      41.5434
53.1829
  std. dev.            9.9068       8.5244       9.4775       7.9795        4.329       7.7144      10.3343
4.6259

job
  unemployed           5.2078      25.1083        7.655      18.3298      21.7263      33.0414           12
12.9314
  services            13.0245      96.1939       2.4811       2.7797     101.0571     138.4599           39
32.0038
  management         174.2534      36.3274      35.1891      558.209      28.8718      17.6222           83
43.5271
  blue-collar         25.3845     162.9629      29.7034       1.4959      85.8441     530.5517           57
61.0576
  self-employed       21.2903      20.8501       6.6413      57.1209      18.6274      39.8668            8
18.6032
  technician          51.2398     150.1321       9.5118      111.265      170.021     166.8445           63
53.9858
  entrepreneur        20.3084      13.9437       4.2105       44.945      10.3429      46.1249           13
23.1245
  admin.              13.8345     120.7053      14.0276      17.0859      98.8949     147.1101           45
29.3417
  student             10.9441      17.9336       1.0452       5.4073      45.0845       1.4673            9
1.118
```

```
  housemaid              14.7557        8.8026       18.1881        8.3643        4.4333       16.7301           18
30.7258
  retired                11.038         1.2674      141.8549        1.8013        3.7506        3.4135           18
56.8743
  unknown                 4.2361        4.1569       14.1195        4.1076        1.3785        3.2995            4
10.7019
  [total]               365.5172      658.3841      284.6275      830.9117      590.0324     1144.5319          369
373.9952

marital
  married               217.1106      388.8173      225.2338      452.7804      128.8335      876.0679          231
285.1564
  single                110.0862      202.9094        4.7033      266.2152      422.6777      101.8263           83
12.5818
  divorced               29.3204       57.6573       45.6904       102.916       29.5212      157.6377           46
67.2569
  [total]               356.5172      649.3841      275.6275      821.9117      581.0324     1135.5319          360
364.9952

education
  primary                27.7649       78.1609      114.3438        3.2826        8.9205      293.5644           54
105.9628
  secondary              86.1892      470.0347       92.5291       50.8897      452.2951      804.8688          169
188.1934
  tertiary              228.1374       68.051        38.3979      755.1105      101.5337        2.1571          121
43.6124
  unknown                15.4257       34.1375       31.3566       13.6289       19.2831       35.9416           17
28.2266
  [total]               357.5172      650.3841      276.6275      822.9117      582.0324     1136.5319          361
365.9952

default
  no                    352.2645      646.5216      271.9552      807.4453      561.9747     1108.9655          351
352.8732
  yes                     3.2527        1.8625        2.6723       13.4664       18.0577       25.5663            8
11.122
  [total]               355.5172      648.3841      274.6275      820.9117      580.0324     1134.5319          359
363.9952

balance
  mean                 8006.6387     2652.2828     2406.6486      678.1413      249.0572       408.304            0
190.9453
  std. dev.            6928.3768     1663.4983     2051.2188      707.2153      299.6285       522.784            0
185.3422

housing
  no                    193.7491      215.9672      235.4842      371.4944      274.6592      256.9523          188
233.6936
  yes                   161.7681      432.4169       39.1434      449.4173      305.3732      877.5796          171
130.3016
  [total]               355.5172      648.3841      274.6275      820.9117      580.0324     1134.5319          359
363.9952

loan
  no                     326.587      573.8412      248.7764      707.4651      476.4142      925.0609          305
274.8552
  yes                    28.9301       74.5429       25.8511      113.4466      103.6183       209.471           54
89.14
  [total]               355.5172      648.3841      274.6275      820.9117      580.0324     1134.5319          359
363.9952

Log likelihood: -15.77713


Time taken to build model (full training data) : 30.25 seconds

=== Model and evaluation on training set ===

Clustered Instances

0      319 (  7%)
1      546 ( 12%)
2      250 (  6%)
3      871 ( 19%)
4      565 ( 12%)
5      1243 ( 27%)
```

```
6       357 (  8%)
7       370 (  8%)
```

Δημιουργήθηκε με το Weka Web (μεταφορά του Weka 3.8.7, University of Waikato, στον browser). Όλοι οι υπολογισμοί έγιναν τοπικά.